P101 · Unit 6

Momentum, work and energy balances

Recall: retrieve · Understand: explain/derive · Consult: look up. Conditions belong with formulas.

Use	Topic	Result and conditions
Recall	**Momentum**	$p = mv$; $P_x = \sum m_i v_{ix}$. Units kg m/s. Common inertial frame; signed components. L1
Recall	**Impulse**	$J_x = \overline{F}_x \Delta t = \Delta p_x$. Units N s = kg m/s. Same interval; constant mass for $\Delta p_x = m(v_x - u_x)$. L1
Understand	**System balance**	$P_{fx} - P_{ix} = J_{\text{ext},x}$. Fixed matter; internal impulses cancel. Negligible external impulse gives approximate conservation. L1
Recall	**Sticking**	$v = \frac{m_A u_A + m_B u_B}{m_A + m_B}$. Positive masses; common final velocity; negligible external impulse. All velocities signed. L2
Understand	**Collision checks**	Momentum alone cannot determine two unknown final velocities. Elastic: total K unchanged. Sticking generally reduces K; total energy is not destroyed. L2
Recall	**Constant-force work**	$W = F \cdot s = Fs\cos\theta$; $W = F_x s_x + F_y s_y$. θ: angle between vectors. J = N m. Particle model; constant force vector. L3
Recall	**Kinetic energy**	$K = \frac{1}{2}mv^2 \geq 0$; $W_{\text{net}} = K_f - K_i$. Add all force works. $v = \sqrt{2K_f / m}$ gives speed, not direction. L3
Understand	**Zero work**	Zero displacement or perpendicular force gives zero work. Zero net work can also mean cancellation of nonzero contributions. L3

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Energy and interpretation

Use	Topic	Result and conditions
Recall	**Uniform gravity**	$\Delta U_g = mg(h_f - h_i) = -W_g$; $U_g = mgh$ for chosen zero. Heights vertical; constant g. Energy belongs to Earth–object interaction. L4
Understand	**Mechanical energy**	$E_{\text{mech}} = K + U_g$; $\Delta E_{\text{mech}} = W_{\text{other}}$. Do not count W_g again. Conserved only if other net work vanishes. L4
Consult	**Explicit ledger**	$K_f = K_i + mg(h_i - h_f) + W_{\text{in}} - E_d$. $E_d \geq 0$: dissipated amount. Do not also subtract the same friction work. Units J. L4
Understand	**Feasibility**	$K_f < 0$ rejects the stated endpoint. $K_f \geq 0$ alone does not check barriers, contact or travel time. Numerical agreement is not experimental validation. L4
Consult	**Method choice**	Force/acceleration: Newton. Force/time: impulse. Force/displacement: work. Collision then climb: separate intervals and assumptions. L5

Consult · Python lookup

```
# SI values; m > 0; inputs finite.
p_i = m_a * u_a + m_b * u_b
v_joined = p_i / (m_a + m_b)
k_i = 0.5 * m * speed_i**2
k_f = k_i + m * g * (h_i - h_f) + work_in - dissipated
if k_f < 0:
    raise ValueError("Impossible endpoint")
speed_f = (2 * k_f / m)**0.5
```

Assign inputs first; the collision and energy snippets describe separate scenarios. For complete input guards use Lessons 2 and 4 functions. Benchmarks: (0.60,1.5,0.40,−0.50) → joined 0.70 m/s; 2 kg, rest, drop 1 m, loss 7.6 J → final 12 J, 3.46 m/s.

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